Quantitative and Qualitative Methods II

Budgeting for Safety:

Linking Defence and Climate Spending in the EU to Security Perceptions in Austria

Annika Müller

Filipa Reis

Hanna Höppler

Jardar Lindaas Bringedal

Karla Röckener

Contents

1 INTRODUCTION 1

2 LITERATURE REVIEW 3

3 METHODS AND METHODOLOGY 8

3.1 Mixed Methods Design 8

3.2 Quantitative Method 9

3.3 Qualitative Method 11

3.3.1 Data selection 12

3.3.2 Data analysis 13

3.4 Integration of Quantitative and Qualitative Methods 15

4 RESULTS 17

4.1 Quantitative Results 17

4.2 Qualitative Results 23

***5* DISCUSSION 29**

5.1 Limitations and Further Considerations 31

6 CONCLUSION 33

1 Introduction

On the 23rd of October 2025, the European Commission declared in their "Defence Readiness Roadmap 2030" (European Commission, 2025a, p. 1):

> By 2030, Europe needs a sufficiently strong European defence posture to credibly deter adversaries and respond to any aggression. To be "2030 ready", Europe needs to move now. [...] The need to speed up and ramp up efforts reflects the increasing dangers of today. [...] Finally, we must contain horizontal threats such as terrorism, proliferation of weapons of mass destruction and the security consequences of climate change.

The roadmap elucidates several measures that the EU will undertake to make the Union defence-ready against "the increasing dangers of today" through increasing EU defence funding, summed up under the *Defence Readiness 2030* initiative. First called *ReArm Europe* but later renamed after criticism from several EU member states (Liboreiro, 2025), the initiatives' objectives remain the same: supporting the EU defence industry, accelerating disruptive defence innovation and ensuring readiness for the worst extreme military contingencies by 2030. The aim is to mobilize €800 billion of both public and private investment by allowing for more deficit spending and by broadening the lending of the European Investment Bank Group (European Commission, n.d.).

The EU is not alone in increasing their military spending: global military expenditure has been rising for a decade now, facing its sharpest increase of 9.4% in 2024, the steepest year-on-year increase since the end of the Cold War (SIPRI, 2025). While this is therefore a global development, Europe, together with the Middle East, has become the forerunner in increasing military spending. This development can be attributed to the current "geopolitical shift" in global power dynamics, exemplified by Russia's invasion of Ukraine in 2022 (Dahal, 2024).

Increases in defence spending are often justified by the "1% principle" commonly used in defence planning, i.e. the principle that some risks have such catastrophic outcomes that even a 1% chance of them occurring necessitates a comprehensive response (Scanlan, 2018). However, such security risks are not unique to major interstate conflict but share the same broad characteristics with climate change. This means that both issues signify a non-quantifiable non-trivial probability of catastrophe (Scanlan, 2018). Still, public climate expenditures are lower than public military spending in a majority of EU member states (Eurostat, 2025). In a similar vein, the European Commission claims that "investing in European defence means investing in lasting peace and long-term stability for both current and future generations" (European Commission, n.d.). Conversely, none of the measures included in the *Readiness 2030* initiative address the "horizontal threat" of the security consequences of climate change mentioned in the roadmap.

While overall climate investment is also increasing globally, this growth is neither sufficient to mitigate global temperature rise, nor is it consistent across sectors and regions

(Climate Policy Initiative, 2025). Public expenditure specifically could prove to be a significant contribution to climate mitigation as there is empirical evidence for negative correlations between public expenditures in the environmental domain and emissions (e.g. Haikos & Paizanos, 2013). At the same time, it has been shown that higher military expenditure jeopardizes reaching the climate targets agreed upon in the 2015 Paris Agreement due to its connection with increasing carbon emissions (Dong et al., 2025).

Studies on defence or climate spending focus on *either* defence *or* environmental spending. Another factor that has so far been overlooked in research on security spending trajectories is public perception, both of federal budgetary decisions and of the security threats themselves. What does the public see as current threats and how does it perceive the budget allocations related to them? Methodologically, macro-level expenditure analyses and micro-level security perceptions remain siloed, disregarding that neither budgetary decisions nor security perceptions evolve independent of one another.

The question we therefore attempt to answer is: What are the dynamics between defence and climate budget spending and public perceptions of security in the EU and Austria? Our work aims to explicitly link spending patterns and threat perceptions, connecting traditional to non-traditional security domains. With a mixed methods approach, we combine a quantitative time-series analysis of EU-wide expenditure data with a qualitative thematic and hermeneutical analysis of a focus group in Austria, including in-depth national-level perceptions.

First, we situate our research within existing literature on expenditure, budget allocation, securitization and threat perception. Then our methodological approach is elucidated, from mixed, to quantitative, and finally to qualitative methods. After presenting the quantitative as well as the qualitative results, the results are interpreted together in our discussion.

2 Literature Review

Analysing budget allocations goes back to the field of public sector economics, the study of a government's economic policy. The research field is usually approached through either the normative or positive theory of the public sector. While the normative theory focuses on what governments should be doing in accordance with societally broadly accepted norms, the positive theory considers the incentives and economic effects of governmental institutions and policies, disregarding accepted norms (Tresch, 2023). Our work will mostly approach the research through the lens of the normative theory of the public sector as we consider public perception of federal economic policy.

Military expenditure

Military and defence expenditure are mostly used synonymously. They include both the expenditures necessary to fulfil the function of national defence and those that arise from fulfilling regional or global security (depending on the state), in addition to those for maintaining internal security (Florișteanu, 2023, p. 197). Drivers of the level of the military expenditure of a state can be grouped into three types: political, military, and economic factors whose influence differs from state to state (Maizels & Nissanke, 1986). Florișteanu (2023, p. 198) identifies further currently relevant factors: geostrategic factors (depending on the level of threat experienced by the state) and new global security challenges characterized by the globalized and diffuse character of the threats or risks, such as cyberwarfare or terrorism.

In the EU specifically, Russia has come to be seen as a potential military threat – even before the Russian invasion of Ukraine in 2022 – resulting in defence spending increases, especially by EU member states sharing land borders with Russia (Christie, 2019). With the Russian invasion of Ukraine, the EU's concept of (open) strategic autonomy, meaning the endeavour to reduce the EU's external dependencies, has regained attention in the EU's policymaking, especially in defence (Miró, 2023). The Commission's *Readiness 2030* initiative, published in 2025, demonstrates how geopolitical threats now impact military expenditures across the EU, highlighting the importance of "level of threat experienced by a state" as explicated in Florișteanu (2023).

Other than research on how higher military expenditure also increases carbon emissions (e.g. Dong et al., 2025) or superficial references to climate change as a security threat (e.g. Florișteanu, 2023), there is limited engagement with environmental threats in the literature about military expenditure, illustrating a narrow conception of security within this research.

Climate security

However, the issue of climate change is increasingly addressed as a security issue (Brzoska, 2009; Arnall, 2023). In Campbell et al. (2009), experts on foreign policy, national security, history and climate science estimate security implications in a scenario with an average global

temperature increase of 1.3°C by 2040 which was already surpassed in 2024 (Copernicus Climate Change Service, 2024). The security implications of this scenario are heightened internal and cross-border tensions (caused by large-scale migrations), conflicts sparked by resource scarcity, increased disease proliferation and as a result of all the above geopolitical reordering.

The meaningfulness of the "climate security rhetoric" is highly discussed (Arnall, 2023). On the one hand, framing climate change as a security threat has increased public attention to the issue. On the other hand, this development might carry the risk that climate change will primarily be addressed via traditional means of security, i.e. by protecting the sovereignty and territorial integrity of a nation (Brzoska, 2009), leading to militarization and the depoliticization of climate change (Arnall, 2023). Brzoska (2009) illustrates that the latter depends on how climate change is conceived: If it is framed as a traditional or "state security" threat, addressing it via traditional security means is more likely. Yet, if climate change is conceived as a "human security" threat, i.e. all threats to human life and well-being, this is not the case. Overall, many governments are hesitant to adopt the rhetoric of climate security (Arnall, 2023, p. 5) but seek to mitigate the security threats of climate change with environmental protection spending.

Environmental protection expenditure

The European Environment Agency (2025) defines environmental protection expenditure as "economic resources used for prevention, reduction and elimination of pollution and any other degradation of the environment". In their analysis of the public environmental protection expenditure of EU member states from 2005 to 2017, Rokicki et al. (2019) found that while it has been systematically growing, the overall significance of such expenditures in national budgets has remained low, not surpassing 2% of the EU's GDP. Differences between the share of protection spending were found to be attributed mostly to the economic situation of the country: Bigger economies tend to invest more in environmental protection, with the overall societal wealth having no impact on the budget share.

Carmona et al. (2023) show based on data from France, Italy, Japan, the United Kingdom and Germany that lower expenditures on environmental protection are causally connected to higher ecological footprints. In contrast, Barrell et al. (2021), in their analysis of the efficiency of environmental protection spending in the EU, point out that environmental outcomes and the efficiency of such expenditure vary among EU member states. This indicates the importance of choosing efficient approaches and strategies for environmental protection and pollution reduction. Furthermore, it also emphasizes to not only consider the amount of expenditure but to also take a closer look at the intended purposes of the budget. Altogether, the environmental protection spending literature is only weakly integrated in broader debates on military and defence expenditure, limiting the understanding of how environmental and military security priorities evolve alongside each other.

Throughout the rest of the paper, we will refer to defence and military expenditure as well as climate and environmental protection expenditure synonymously.

Economic budgetary process and trade-offs

The overall process of budget allocation or "budget-making" is both an economic as well as a political process. Economic budget-making can be seen as the most efficient budget allocation of limited resources that helps governments reach certain objectives (Tresch, 2023). Since government resources are limited, budgetary trade-offs among different areas of fiscal policy are bound to happen (Looney, 1986), meaning that an increase in one fiscal area could decrease or "crowd out" expenditures in other fiscal area (Günluk-Senesen, 2002, p. 387). These budgetary trade-offs have mostly been studied between military and social spending in the course of the so-called "Guns vs. Butter" debate, showing that e.g. rising public welfare expenditure decreases military spending (Carter et al., 2020).

The findings on trade-off relationships between military expenditure and other fiscal areas are difficult to generalise. First, they are hardly comparable because of model and data differences. Second, they also diverge in terms of even identifying the existence of trade-off relationships and in terms of which fiscal areas are adversely affected (Günluk-Senesen, 2002, p. 388). Third, governments can also utilize debt, growth or other kinds of fiscal flexibility to finance budget increases and therefore diverge in the limited nature of their fiscal resources. Relationships between military and climate expenditure specifically have not yet been explored.

The Stability and Growth Pact of the 1992 Maastricht Treaty obligates EU member states to avoid excessive deficits in their expenditure (Schuknecht et al., 2011), disincentivizing deficit spending and thus creating the conditions for budgetary trade-offs between different fiscal sectors up until 2025. E.g. Austria violated the terms of the Pact as their "general government Maastricht balance" amounted to –4.7% of GDP, surpassing the threshold of –3.0% of GDP and thus initiating the EU's Excessive Deficit Procedure. The Austrian federal government is now "complying with the requirement to present effective measures to correct the excessive deficit" (Austrian Federal Ministry of Finance, 2025), meaning it instituted austerity measures. This exemplifies the economic conditions for budgetary trade-off relations in the EU.

With the implementation of the *Readiness 2030* initiative, the Commission proposed an exception for defence spending from the Stability and Growth Pact, allowing for additional spending in this sector up to 1.5% of GDP (European Commission, 2025b). This will likely distort future trade-offs between military expenditure and the other fiscal sectors as member states can increase this budget without decreasing the others.

Political budgetary process and political agenda setting

Wildavsky (1964, p. 4) explains why "budget-making" is also a political process: "If politics is regarded in part as conflict over whose preferences shall prevail in the determination of national policy, then budget records the outcome of this struggle." The allocation of resources in the budgetary process is seen as reflecting both the distribution of power and the norms people live by in the respective society – in line with the normative theory of the public sector (Tresch, 2023).

Within this political process of budget allocation, what is on the political agenda matters. The concept of political agenda setting based on the seminal work of Schattschneider (1960) helps to understand how issues ascend and descend on the political agenda, and what policy consequences this prioritization has. Mortensen (2010) specifically shows that the amount of attention given to an issue by national policy makers impacts the allocation of public spending. The direction of this effect – if it increases or decreases the budget of an area of fiscal policy – is determined by the spending attitudes of the mass public. This is suspected to be due to the re-election-oriented behaviour of politicians.

The notion of "issue competition" (Budge & Farlie, 1983) furthermore emphasizes the multitude of factors that affect what the political agenda looks like as it states that in party competition, parties try to draw attention to issues they find advantageous. Green-Pedersen and Mortensen (2009) e.g. find that opposition parties have more opportunities to focus on "advantageous issues" and due to controlling party-system agenda can "force" government parties to address these.

Taken together, this literature illustrates that government expenditure decisions are not taken purely due to economic efficiency considerations but also reflect a deeply political choice process: Fiscal areas get uneven amounts of attention by policy makers, public opinions determine the direction of budgetary changes and parties wield different concerns in the political competition of multi-party election systems. This illustrates how security shocks such as the Russian invasion of Ukraine in 2022 can shift attention of policy makers as well as public opinions on which issues are considered important and need to be addressed, which in turn can influence how fiscal resources are allocated.

Securitization and threat perception

What is perceived as a security issue does not necessarily reproduce the objective and material circumstances of the world, as Balzacq et al. (2016) explicate. Securitization theory seeks to conceptualize how public problems are established as security issues and what societal as well as policy consequences this establishment carries, meaning that securitization is a combination of the politics of threat design and of threat management. Securitization occurs when an issue (e.g. an armed conflict) acquires enough saliency to win the assent of the audience (e.g. the public). This then enables entities authorized to handle the issue (e.g. the federal government) to use whatever means deemed most appropriate by those entities. The idea of securitization is based on the notion that "the word 'security' has a performative character – that is, it does not only describe the world but can also transform social reality" (Balzacq, 2016, p. 495).

If an armed conflict is deemed a threat to society by the public, governments can "earn" the authorization to adequately prepare the defence of said society by whatever means thought to be necessary by said government. The same process would be required to securitize the issue of climate change. What is essential in securitization to initiate the process is for the issue to acquire enough saliency, meaning it is deemed important or threatening enough, by the *audience*. It must sanction the securitizing move (e.g. allocating more money

towards military or environmental protection expenditure) and supply the actor with a formal mandate to do so (e.g. voting by the legislature) (Balzacq, 2016, p. 500).

However, threats are rarely perceived in isolation of one another. While many may state high concern for climate change, when people are forced to evaluate the severity of environmental threats in relation to other threats (e.g. external security threats such as armed conflict), environmental threats are ranked at the bottom of the list. This phenomenon is attributed the perceived temporal proximity of the threat: Environmental threats are more temporally distant and thus deemed less severe than external security threats which are ranked at top priority. Individuals with high future orientation, i.e. with the tendency to take into account temporally distant considerations, tend to rank environmental threats higher. Nevertheless, individual high future orientation has no effect on the high ranking of external security threats (Carmi & Bartal, 2013).

What is deemed a security threat is therefore socially constructed and if an issue is securitized this entails significant consequences. In the world of the polycrisis, with growing complexity and interactions among global crises (Rakowski et al., 2025), threats compete for attention and their severity is evaluated in relation to one another. Thus, if a “new” threat emerges such as Russia due to its invasion to Ukraine, this may displace “old” temporally distant threats such as climate change from the priority list of the public.

Methodological gap

The existing literature on military and environmental government expenditure and public security perceptions appears to show a number of empirical and conceptual gaps. First, macro-level quantitative studies on government expenditures do not connect defence and environmental protection spending. Second, the literature largely neglects the integration of proxies for public perceptions of security, thereby also neglecting one of the drivers of budget allocation, i.e. public opinions. Lastly, studies on securitization and threat perception are rarely connected to actual policy outputs and fiscal expenditure.

By exploring the evolution of fiscal priorities in the EU and Austria and linking it to the social construction and perception of security threats of EU-citizens in Austria through a mixed methods approach, our research aims to fill a gap in the existing literature.

3 Methods and Methodology

3.1 Mixed Methods Design

Mixed methods is a methodology that combines quantitative and qualitative research. It assumes that both quantitative and qualitative methods provide different information about complex phenomena. Hence, combining both approaches expands our understanding. Ideally, the strengths of both approaches complement one another, whereas their limitations are mitigated (Creswell, 2009; Creswell & Creswell, 2018). Mixed methods approach add complexity by combining two or more methods with distinctive ontological and epistemological principles (Bryman, 2012). Yet, the clear advantage lies in the triangulation of methods, "the idea that looking at something from multiple points of view improves accuracy" (Neuman, 2014, p. 166).

The timing, weight, and integration of quantitative and qualitative methods are important considerations in mixed methods designs (Creswell, 2009). The timing of our quantitative and qualitative data collection and analysis was predominantly concurrent. The differing time requirements of the quantitative and qualitative methods led to an earlier quantitative analysis, while the qualitative component required additional time for data collection and analysis. We split into two research groups: a quantitative and a qualitative team. Despite the split-up, regular update meetings between the quantitative and qualitative teams were held. The weight of our quantitative and qualitative research was equally distributed. Yet, during our research, primary quantitative data informed the input used for the selected qualitative method. Later, our qualitative results added value to the quantitative results in the discussion. The integration of both research methods occurred during the planning of our research design, the analysis, and especially the discussion and interpretation of our results.

A variety of mixed methods designs have been developed in the literature. Due to the aspects of timing, weight, and integration, a hybrid approach combining a triangulation and an explanatory design was employed in our research project. In a convergent triangulation design, quantitative and qualitative data collection and analysis occur simultaneously, underscoring the equal weight of both methods. The mixing arises in the discussion of the results. In an explanatory design, there are two phases of data collection and data analysis. The quantitative results inform the qualitative research, or vice versa, thereby assigning different weights to the methods. The methods are mixed, with one method enhancing the results of the other (Creswell, 2009; Creswell & Creswell, 2018). Our hybrid approach combined the two designs in a way that balanced their respective strengths and weaknesses.

The hybrid design choice was informed by multiple considerations. First, compared with an exclusively explanatory design, in which one method builds on the other, the hybrid design is more time efficient. Given that the research was a group project within a limited timeframe, it was beneficial to divide the tasks in our research teams and work concurrently on both methods. Second, a limitation of an exclusively triangulation design is the assumption

that both quantitative and qualitative methods address the same variables and themes, yielding comparable results (Creswell, 2009; Creswell & Creswell, 2018). However, the hybrid design approach enabled us i.e. to address the theme of sense of security in our qualitative research, thereby enhancing our understanding of the dynamics between budget spending and public perceptions of security.

Mixing quantitative and qualitative methods provided a more comprehensive answer to our overall research question. The quantitative methods addressed research gaps at the EU-level, whereas the qualitative methods offered an in-depth view of the Austrian case. Furthermore, quantitative methods are prone to a static picture, compared to more processual qualitative methods. Here, our qualitative research offset the limitation of our quantitative research, which can only represent fixed past processes, and enables a processual analysis that captures current changes in policy and zeitgeist toward militarization (Bryman, 2012).

3.2 Quantitative Method

The quantitative method is an analysis of time series models of defence budget and environmental protection budget in the 27 EU member states plus Iceland, Norway and Switzerland over the period from 1998 to 2023. These data are retrieved from Eurostat and is called "General government expenditure by function" with the code (gov_10a_exp) (Eurostat, 2026). The Eurostat database was used due to feasibility, with easily accessible and comparable data across the 30 investigated countries.

Inside the defence budget there are five subcategories: military defence, civil defence, foreign military aid, Research and Development (R&D) defence and defence n.e.c. (not elsewhere classified). Military defence most of the time represents around 90% of the total defence budget (Eurostat, 2026). The environmental protection budget consists of six subcategories: waste management, water waste management, pollution abatement, protection of biodiversity and landscape, R&D environmental protection and environmental protection n.e.c. (Eurostat, 2026). We decided to modify the environmental protection budget for our data analysis, withdrawing waste management and water waste management, while keeping the four other subcategories. In this regard we diverge from the European Environmental Agency's (2025) definition of environmental protection expenditure as "economic resources used for prevention, reduction and elimination of pollution and any other degradation of the environment". In the analysis waste management and water waste management was seen as a minimum of what is needed regarding cleaning up waste from the past. The remaining four subcategories, pollution abatement, protection of biodiversity and landscape, R&D environmental protection and environmental protection n.e.c. are seen as active investments in trying to reduce and prevent future degradation of the environment and not only keeping up a status quo like waste management and water waste management.

Instead of using absolute amounts in €, percentage spent of GDP is employed in the analysis, the main reason being the high inflation post Covid-19 that caused large growth in

expenses. The start of our quantitative analysis is a descriptive statistical exploration of the data, including plots of the budget allocation and spatial plot of the budget trends in the last 10 years.

The first quantitative research question is: **How have climate and defence budget allocations in the EU and Austria evolved between 2019 and 2023?** To answer this question, we decided to execute an autoregressive integrated moving average known as an ARIMA analysis of our dataset, which is a type of times series analysis. To employ the ARIMA model there are six steps that Shumway and Stoffer (2016) recommend, starting with plotting the data. After downloading the budget data from Eurostat, the data are cleaned and transformed to conform to the analytical setup. Then the dependence orders of the variables are identified. The next step is to find the values of the autoregressive order (p), the order of differencing (d), and the moving average order (q). These are also known as ARIMA (p, d, q). ARIMA models must be stationary and the way to obtain this is by differencing the time series to obtain stationarity. The number of times of differencing decides the d-value (Coghlan, 2026). The p-value is the number of steps into the past the autoregressive model uses to forecast current values (Shumway & Stoffer 2016). The q-value uses current and (q-times) past noise terms to determine current value. The p and q values can be checked by making partial correlogram for p and correlogram for q (Coghlan, 2026). Then we can create a diagnosis that leads to the final step of choosing which ARIMA model, meaning which p-, d-, q-values fit our data. Out of feasibility reasons we did not create a specific ARIMA (p, d, q) for each of the 30 countries for both budget types, rather deciding on ARIMA (1, 2, 2) for the environmental protection budget and ARIMA (1, 1, 2) for the defence budget after trying out different values.

After obtaining the ARIMA with the q, d, p values it is possible to use this to forecast the budget development. Looking at the initial data and time series plots before conducting the ARIMA model we saw no big change in either of the two budgets on average over the 30 countries. However, for some countries, there is a steep increase for either one or both of the budgets over the last years. To answer the research question, we decided to forecast the years from 2019 to 2023, using the dataset until 2019 to predict each country's budget developments, seeing if any of the countries had unexpected developments. The intention behind examining data only until 2019 was to exclude the budget changes due to the Covid-19 pandemic before measuring in how far Russia's invasion in Ukraine affected budget developments.

Based on the real development until 2019, the ARIMA model forecast how the spending would develop until 2023. Then the forecast was compared to the real budget data for 2023. If the forecasted trend and the real 2023 data were the same, then one cannot see any effect of the 2022 invasion in the data frame. If it differs, then the 2022 invasion might have had an influence on the budget developments.

As a second step, we want to answer our second quantitative research question: **Is there a trade-off between climate and defence budget spendings in the EU and Austria?** We analyse the factors associated with changes in EU government public expenditures, by

estimating two Ordinary Least Square (OLS) regressions, using the cross-sectional data for European countries. In each of the OLS, the dependent variables are the differences between actual values for 2023 and the ARIMA forecast of Defence and Environmental Protection Budget expenditures, thus capturing spending changes not explained by historical trends.

M 1: $diff_{env_i} = \beta 0 + \beta 1\, diff_{def_i} + \beta 2\, \log(distcap_i) + \beta 3\, nato_i + \beta 4\, debt_i + \beta 5\, energy_i + \varepsilon_i$

M 2: $diff_{def_i} = \beta 0 + \beta 1\, diff_{env_i} + \beta 2\, \log(distcap_i) + \beta 3\, nato_i + \beta 4\, debt_i + \beta 5\, energy_i + \varepsilon_i$

The main explanatory variables in both models include the difference between actual and forecasted values of the other spending category, the logarithm of the distance of each country to Russia (capital distances) as a proxy for geopolitical risk exposure, NATO membership in 2023 (excluding Finland and Sweden) (NATO, 2024), government debt-to-GDP ratio and country energy intensity. The models are estimated using OLS due to the continuous nature of the dependent variables, the cross-sectional structure of the data. Coefficients are thus interpreted as the marginal association between each explanatory variable and the magnitude of the deviation from forecasted spending, ceteris paribus. As such, statistically significant coefficients indicate systematic cross-country patterns in policy response beyond what would be expected based on past expenditure dynamics alone.

All data wrangling, visualisations, descriptive statistics, forecasting and econometric analysis were implemented in R and are fully documented in a publicly available GitHub[1] repository. The repository contains the complete workflow, including all the specifications and generations of tables and figures, as well as all the original datasets used. Providing a full code ensures transparency, reproducibility and facilitates further extensions and robustness checks of the analysis.

3.3 Qualitative Method

The aim of the qualitative part is to dive deeper into the case of Austria and explore how changes in the defence and climate budgets are affecting people's sense of security. With regards to content, this research can be situated within the normative theory of public sector economics, as set out in the literature review, which focuses on how government action aligns with societally accepted norms (Tresch, 2023). If budget allocation is understood as a political process that reflects, first, the distribution of power and, second, societal norms, a closer examination of public perceptions provides a better understanding of the latter. Accordingly, the qualitative part offers an in-depth perspective on Austrian societal norms and public opinions on climate and defence budgets. In addition, given the lack of direct democratic participation in NATO and EU defence-spending directives and limited direct participation in

[1] https://github.com/filipareis9/QQM_Quanti

budgetary decision-making in Austria, engaging with public perceptions allowed us to examine this potential disconnect in greater depth.

Specifically, this part of our project will be guided by the following two research questions:

- **How do Austrian inhabitants perceive climate and defence budget shifts in relation to their sense of security?**
- **How are the security topics of climate change and military conflicts put into relation?**

3.3.1 Data selection

Our research questions were approached through a focus group discussion. The focus group is a particularly insightful and substantive method, as it brings together different perspectives, understandings, and opinions on the topic. Furthermore, the setting allows direct observation of how these two topics are approached, put into relation with each other, and negotiated in a social context (Bryman, 2012). The process of "sharing and comparing" among participants enables a deeper understanding of their views and the reasoning behind them, as they are challenged to reflect and respond to differing perspectives (Bryman, 2012).

The sampling of our focus group participants was conducted using a mix of convenience and snowball sampling. We aimed to include individuals living in Austria, not necessarily Austrian citizens, with diverse backgrounds in terms of education, gender, age, and other characteristics. However, as is typical for qualitative research, this study does not aim to provide a representative picture of society as a whole, but rather to offer in-depth insights into social phenomena. Nevertheless, we sought to select participants who reflect a range of perspectives on the topic, including those more supportive of climate-related spending and those more favourable towards military expenditures, through recruitment in diverse social contexts. Participants were mobilized, among other strategies, through the distribution of flyers. Ultimately, the participants in our focus group represented a range of perspectives but were generally relatively more supportive of climate-related spending and well educated on both topics of climate change and militarization, resulting in more homogeneous outcomes than anticipated.

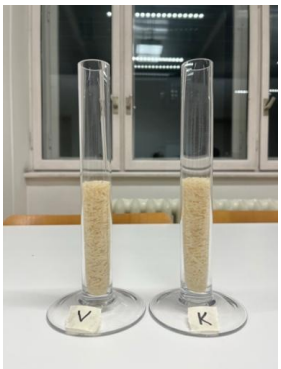


Figure 1: An exemplary result of the individual budget allocation exercise. V stands for defence budget, K for climate budget.

Our focus group discussion lasted two hours and included six participants. The discussion was structured as follows: To guide the discussion around the two security themes of climate change and military conflict specifically, we began with a brief introductory input that was deliberately kept minimal to not induce excessive prior framing of the topic. Afterwards, the approach consisted of intervening as little as necessary to allow for a free flow of discussion and the emergence of topics driven by the participants themselves. Yet we used a set of guiding questions covering key themes that were aimed to be addressed

during the discussion, if not brought up by the participants themselves. The session was further complemented by interactive elements, which were primarily intended to stimulate further discussion and refocus the group on the research topic and guiding questions. These included a budget allocation exercise in which participants distributed hypothetical expenditures individually and collectively by pouring rice into designated glasses, as well as an exercise where participants were asked to rank images depicting security threats related to climate change or military conflicts by their sense of (in-)security.

3.3.2 Data analysis

Our data analysis consisted of a combination of two methods: hermeneutic analysis and thematic analysis. The whole process and the interplay between the methods are illustrated by Figure 2. Hermeneutic analysis allowed us to go beyond the surface level and reconstruct latent meanings, different argumentative structures, and the conceptualisation of actors in greater depth, enabling analytical generalization (Reiter & Sardadvar, 2025). Through this, we were able to gain deeper insights into how, for example, particular security perceptions were defined. In addition, thematic analysis was employed, as it allowed us to systematically identify, organize, and interpret recurring patterns of meaning (themes) within textual data (Braun & Clarke, 2006). The method enabled us to identify which themes emerged in the discussion, for example how and which events and developments at the societal level were brought up by the focus group participants. Consequently, at an analytical level, we examined how these themes were connected to one another.

Our ontological and epistemological orientation that underpins the data analysis corresponds to a critical realist and a weak social constructivist approach. Both approaches argue that a mind-independent reality exists (such as changes in our global climate) but that our knowledge and interpretations of this reality are socially, culturally, and historically constructed. This philosophical stance aligns well with our focus group approach, as it provides a setting where meanings are co-constructed and negotiated among participants. As “security” as a concept can have different meanings for each person, and it is difficult or impossible to define a single objective truth, this perspective allows us to explore and gather various perspectives on how participants interpret and relate budgetary changes to their own sense of security.

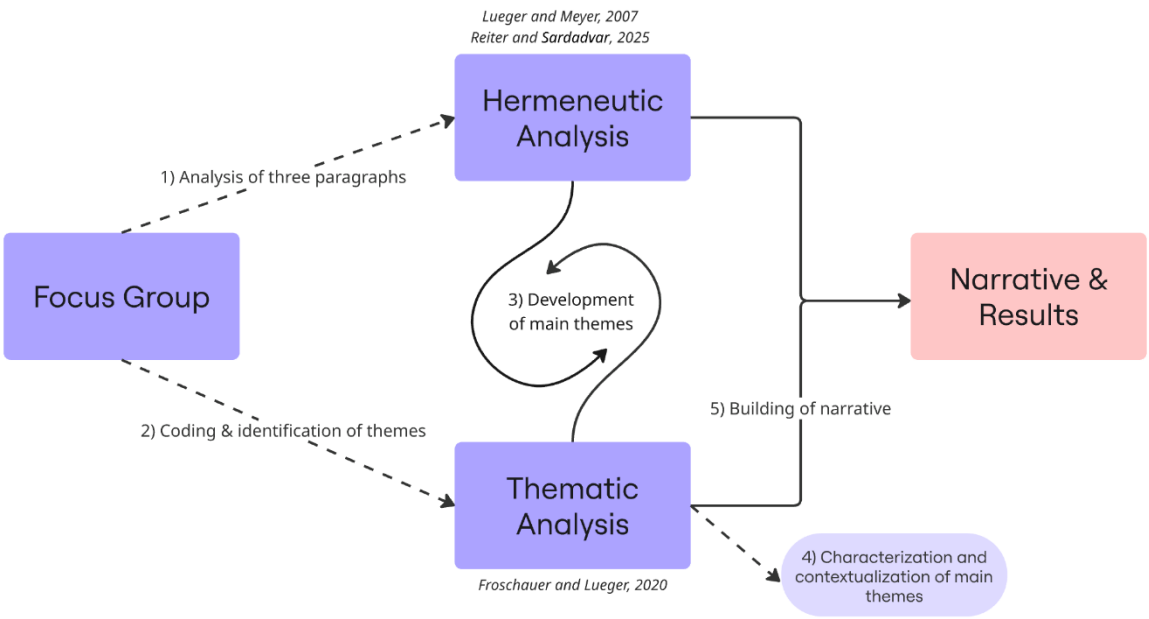


Figure 2: Qualitative analysis process.

Hermeneutic analysis

We began our data analysis with a hermeneutic approach to minimise the influence of prior assumptions and to enable analysis with a relatively open interpretive stance. In total, we selected three passages that appeared to contain substantial latent meaning and narrative depth, and that brought together multiple thematic strands. We conducted the hermeneutic analysis by dividing the selected passages into smaller meaning units to carry out a sequential analysis (Reiter & Sardadvar, 2025).

Subsequently, we continued the analysis using a matrix developed for hermeneutic interpretation, primarily based on Lueger and Meyer (2007). This process involved several steps. First, we analyzed the literal and precise meanings of the words used. Second, we conducted what is described in the literature as an extensive interpretation (Froschauer & Lueger, 2020) which consists of developing as many interpretations as possible. Third, we paid particular attention to the distribution of roles and the characteristics of the actors that emerged in the material. As we conducted the hermeneutic analysis as a team, the process also provided a form of quality assurance in identifying the most convincing interpretations, as it required justification among differing understandings within the group and encouraged extensive reflexivity towards the material (Reiter & Sardadvar, 2025). This was done by individually developing an initial hermeneutic analysis of a given paragraph, followed by a thorough team discussion in which the analysis was further refined and consolidated.

Adding thematic analysis

After conducting the hermeneutic analysis, we proceeded with the thematic analysis, which was based on the approach outlined by Froschauer and Lueger (2020). The identification of themes as a first step of analysis was based on inductive coding to ensure that the themes were grounded in the data. The transcript was coded jointly and simultaneously from the outset. A preliminary characterisation of the themes already emerged during the coding process, as we continuously discussed why particular passages corresponded to specific codes. After gathering and organising the themes, the next step of this approach would usually consist of a detailed characterisation and contextualisation of the themes. However, at this stage, we returned to the hermeneutic analysis. Through our hermeneutic analysis, we already grasped which themes and actors involved notable tensions and insights and which should be included in our main themes. Subsequently, we combined and connected these main themes from the hermeneutic analysis with selected themes and findings from the thematic analysis that we considered particularly relevant for the contextualisation of the hermeneutical results and for addressing our research question. This step already represented our first attempt to gather our qualitative findings from the hermeneutic and thematic analyses in a coherent narrative. This was primarily done through developing a visual mind map that structured the different themes in a logical order and outlined their connections (see Appendix).

Afterwards, we conducted a further detailed thematic analysis of our most important themes in combination with a more condensed analysis for some subthemes. This further analysis involved the characterisation of the themes, their dimensions and different manifestations, as well as further analysis of their social, temporal and factual contextualisation, including, among other steps, identifying how different topics were linked to one another (Froschauer & Lueger, 2020).

Finally, we brought together all themes and findings from our hermeneutic and thematic analyses by further developing and structuring our narrative mind map and by writing them up in a final comprehensive narrative. The connections between topics, which were analysed during the contextualisation of the themes, formed the basis for developing an analytical thread. Furthermore, it is important to note that we translated our findings from German into English at the stage of writing the narrative, in order to remain as close as possible to the original content during the analysis phase and to avoid losing nuances in meaning.

3.4 Integration of Quantitative and Qualitative Methods

The implementation of the research design and the application of methods followed a collaborative, structured mixed-methods workflow, with the integration of quantitative and qualitative methods occurring at multiple stages of the research process.

Collectively, all members of the research team conducted a preliminary literature review and data search to familiarize themselves with the topic, current approaches, and state-of-the-art

methodologies, resulting in a basic analytical framework and research design. At this point, we divided into two teams: one focused on quantitative analysis and the other on qualitative analysis. The decision on member allocation was based on language criteria, as German was required for the implementation of the focus group. The quantitative analysis focused on past budgetary changes on EU level from, whereas the qualitative analysis focused on Austria as a case study. This division enabled both teams to develop their data collection and analysis in parallel, in accordance with a convergent triangulation design, and made our research design feasible given the time constraints.

Regular feedback meetings were held between the two groups to reflect on differing ontological and epistemological perspectives, share interim results, discuss emerging themes, and how insights from one method could inform the other, and vice versa. One obstacle we faced was integrating security into the quantitative analysis to ensure comparable results between the quantitative and qualitative research. However, due to the lack of usable data, security was not included as an additional variable in the quantitative analysis. Instead, the sense of security became the primary focus of the qualitative research, leading to a more explanatory leaning design. Regarding the emerging themes, following the first interim results of the held focus group, we found that the Russian-Ukrainian war was crucial to both quantitative and qualitative analyses. Within each team, additional feedback sessions were organised to discuss methodological choices, coding strategies, model specifications, and practical challenges such as data limitations and interpretation issues. This iterative exchange with both the within-team and the other analysis team enabled continuous refinement of both qualitative and quantitative analyses. In the final stage of the research project, the results from both research strands were brought together in the following discussion section. We aspired to a comparative interpretation of the findings and a critical reflection on the difficulties, methodological limitations, and the added value of combining the aforementioned perspectives. These processes ensured that integration occurred not only at the level of results but also throughout the research process itself.

4 Results

4.1 Quantitative Results

Descriptive Statistics: Budget trends across the EU

To understand changes in expenditure levels, budget compositions, cross-country patterns, and how policy priorities have evolved since the late 1990's, we chose an exploratory analysis of budget expenditure data for the selected European countries during a time span of 25 years, from 1998 until 2023.

Figure 3: Development of defence expenditure, environmental protection expenditure and environmental protection expenditure without (both) waste management, 1998 to 2023

This analysis reveals substantial variation in defence and environmental related public spending, with consistent higher defence budget across the majority of European countries,

the exception being Iceland. Iceland has no armed forces but are still part of the NATO (NATO, 2025b). Most countries show stable expenditures across the years, with countries like Romania, Greece, Latvia, Lithuania, Estonia, Bulgaria and Cyprus showing higher volatility in defence spending.

For each country, changes in both budget expenditures were calculated between 2013 and 2023 as a share of GDP, capturing spatial distributions of medium-term shifts in budgetary priorities.

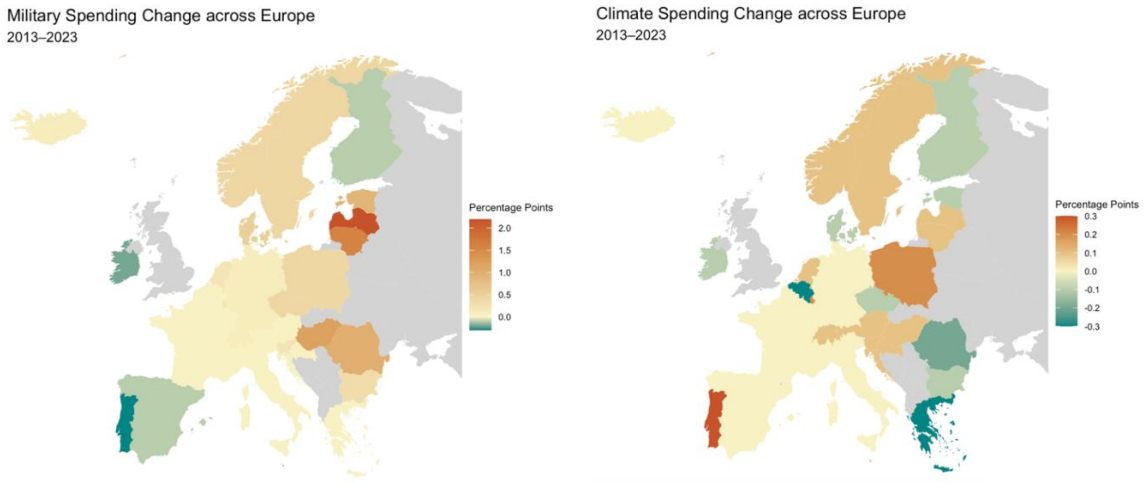


Figure 4: Spatial distribution of military and climate spending changes across Europe, 2013 to 2023 (different scaling for clarity)

Defence spending changes shows the largest increases are concentrated in Eastern Europe, particularly the Baltic countries, while Central Europe generally exhibit stable spending levels and Southern Europe and Finland showing decreases over the past decade. Changes in environmental spending show a more pronounced, fragmented and less geographically coherent pattern. While some countries display moderate increases in environmental spending, namely Portugal, Sweden, Norway and some Eastern European countries, other Eastern European countries, show a stark decline, with no clear regional clustering. This suggests that changes in environmental spending remain primarily driven by country-specific policy choices rather than a shared regional dynamic. In general, both maps show greater volatility in both budget expenditures in Southern and especially in Eastern European countries.

Descriptive Statistics: Defence and environmental budget breakdowns

Taking a closer look at the composition of defence and environmental protection expenditure in the EU-27 countries and Austria, very similar trends are visible. At the EU-level, military expenditure dominates the overall structure of defence budget throughout the 25 years. Since the 2010s, foreign military aid increased and stabilised while comparatively smaller to the other components, and other components being negligible contributions. Austria shows

consistently lower military and defence spending as a share of GDP than the EU-27 figures, with a decline in early 2010s. In the breakdown of environmental spending composition, waste and water waste management are the categories that contribute most to environmental protection budgets. Austria shows a similar structure to that of the EU-27 but with greater volatility across categories, particularly in waste management. Unlike defence spending, environmental related expenditure does not exhibit a clear upward shift in the later years of the analysis period, rather spending levels appear relatively stable.

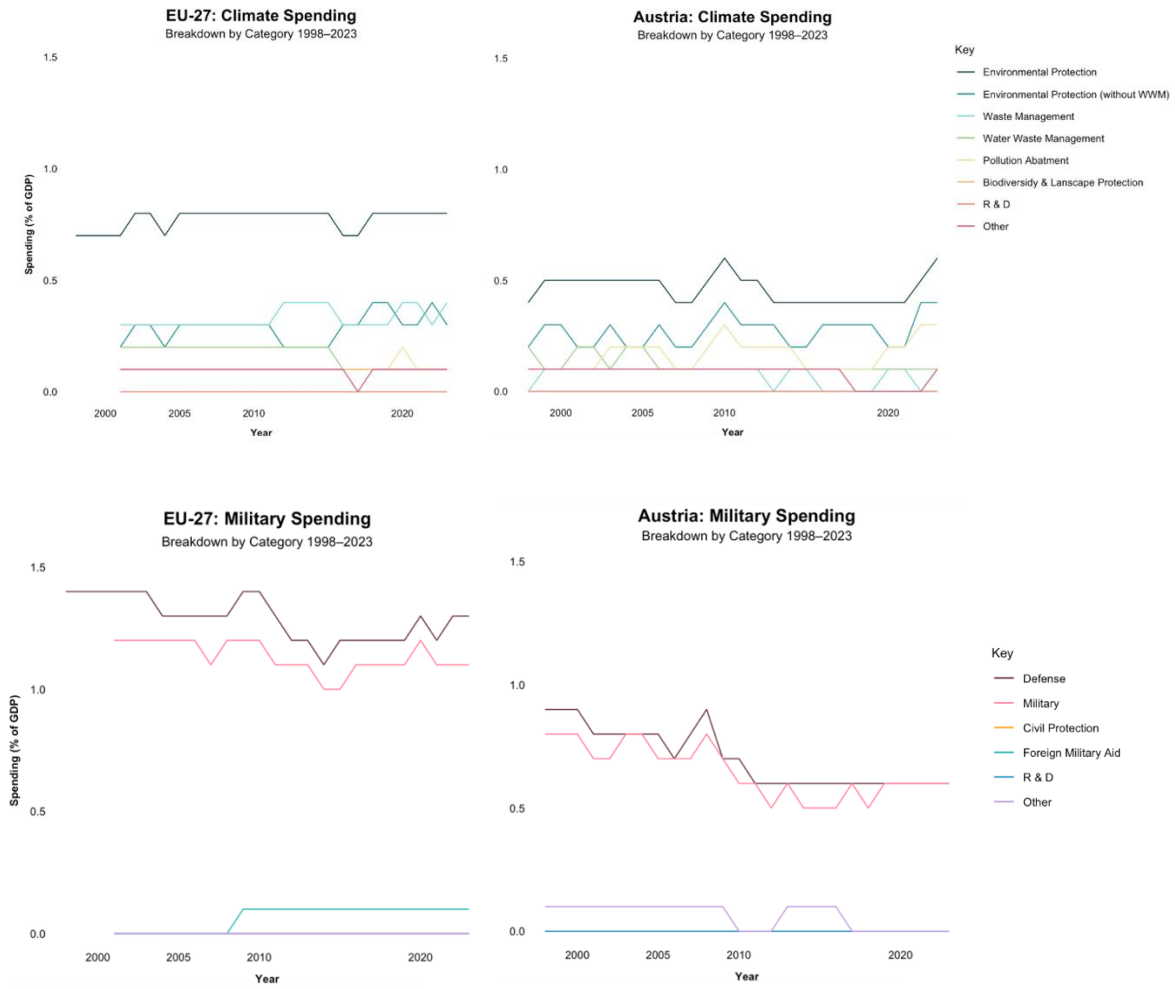


Figure 5: Breakdowns of military and environmental protection spending in the EU-27 and Austria, 1998 to 2023

Results of the ARIMA model on defence expenditure

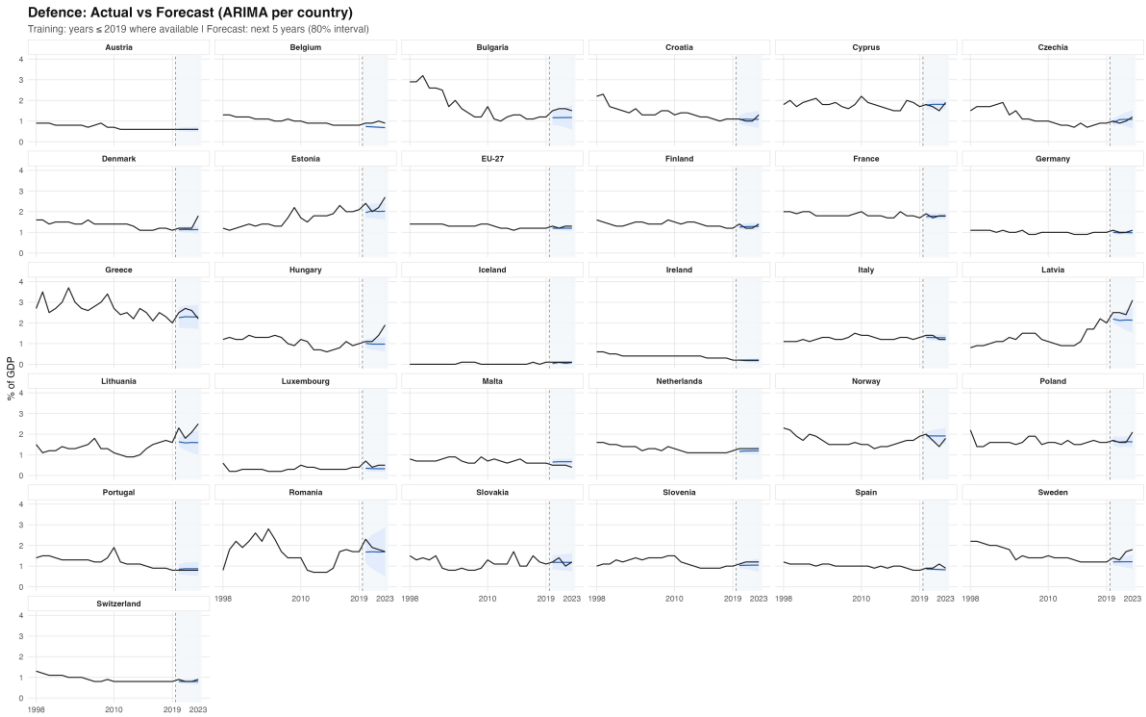


Figure 6: Results of the ARIMA (1, 1, 2) model for defence expenditure

The ARIMA (1, 1, 2) model for the defence budget shows that the EU-27 average has a small increase in the defence budget, but within the 80% confidence interval, meaning there is no statistical significance. This is the case with most of the analysed countries, except for Denmark, Estonia, Hungary, Latvia, Lithuania, Poland and Sweden. All these countries have a bigger growth in the defence budget than the ARIMA forecast expected. The difference is outside the 80% confidence interval for all of them, making these findings statistically significant.

Within these countries are two categories: In Denmark, Hungary, Poland and Sweden the defence budget growth occurs mainly from 2022 until 2023. For most of these countries, the Russian invasion in Ukraine is the strongest factor. In the Baltic countries Estonia, Latvia and Lithuania, budget changes already start in 2015, the year after Russia invaded Crimea.

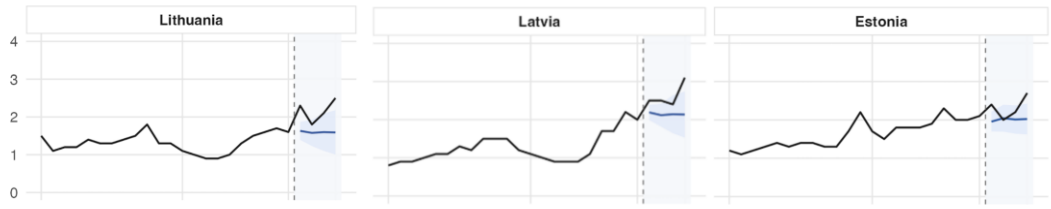


Figure 7: Results of the ARIMA (1, 1, 2) model for defence expenditure for Lithuania, Lativia and Estonia

Results of the ARIMA model on environmental protection expenditure

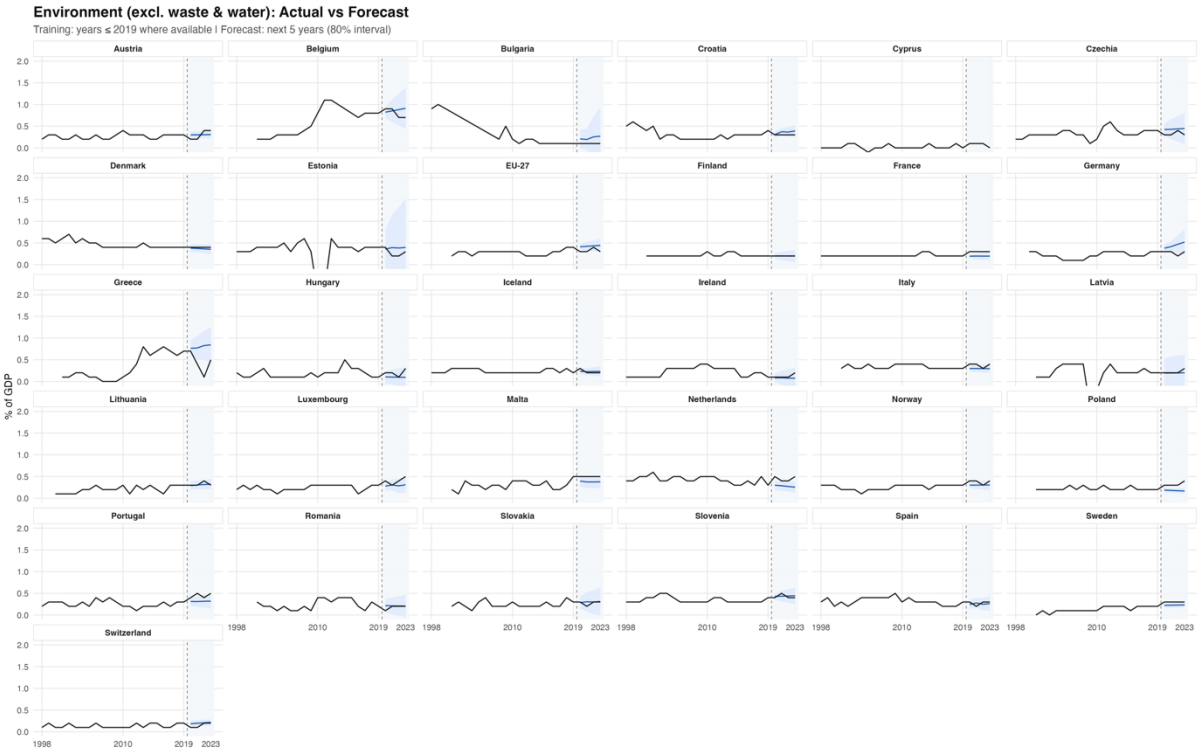


Figure 8: Results of the ARIMA (1, 2, 2) model for environmental protection expenditure

The ARIMA (1, 2, 2) model for the environmental protection budget for the EU average shows a higher forecasted value than the real value of the environmental protection budget in 2023. But the result is within the confidence interval and thus not statistically significant. There are a few countries that have lower forecasted environmental protection budgets than their actual real value in opposition to the EU average. This list contains Luxembourg, Netherlands, Poland and Portugal, all with values outside the 80% confidence interval, making them statistically significant. In the graphs there was no forecast values for Cyprus in the environmental protection budget. Looking into this shows that the value of the Cyprus environmental protection budget was 0 in 2019 and also 0 in 2017. Therefore, the ARIMA (1, 2, 2) model values did not have any values to analyse from and the whole ARIMA (1, 2, 2) model thus failed for Cyprus.

Relationship between defence and environment budget allocations

We estimated two OLS regression models to examine the relationship between the forecast errors in environmental and defence spendings across the selected European countries in 2023. Forecast error is defined as the difference of the observed spending and the forecasted government expenditures. This way, both models are designed to test whether deviations in one policy domain are associations with deviations in the other, controlling for geopolitical risk exposure, NATO membership, fiscal capacity and energy context.

	Model 1: Environmental Spending	Model 2: Defense Spending
(Intercept)	−1.376	4.802***
	(0.897)	(1.089)
diff_def	0.189	
	(0.134)	
log(distcap)	0.207+	−0.607***
	(0.112)	(0.141)
nato	−0.095	0.213
	(0.081)	(0.128)
debt	−0.001	−0.001
	(0.001)	(0.002)
energy	−0.000	−0.002
	(0.001)	(0.002)
diff_env		0.503
		(0.355)
Num.Obs.	25	25
R2	0.282	0.647
RMSE	0.12	0.20

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 8: Results of the OLS estimations for Model 1 and Model 2

Model 1 examines whether differences in defence spending forecasts explain environmental spending forecast differences in 2023. The results suggest that environmental spending discrepancies are only weakly related to the explanatory variables. The estimated relationship between defence and environmental spending discrepancies is positive, meaning that countries with higher discrepancies in defence spending tend to have higher discrepancies in environmental spending. The only variable approaching statistical relevance is the geographical distance to Russia: the positive coefficient suggests that countries further away tend to show slightly larger environmental forecast discrepancies, although this effect is modest and not significant. Overall, this model explains a limited share of variation in the model, which implies that most of the differences between projected and actual spending likely stem from country specific or short-term influences not captured by the model.

Model 2 reverses the relationship, examining whether environmental forecasting errors explain defence forecasting error. While this model explains a substantially larger portion of the variation, which suggest a higher explanatory power of the model for defence forecast deviations, environmental forecasting differences do not significantly explain defence forecasting differences, reinforcing the ideas that these budgeting processes operate largely independently. However, distance to Russia shows a statistically significant and negative association with the spending forecast discrepancy, which implies that relatively shorter geographical distance is associated with larger defence forecast differences - an observation also supported by the spatial trends initially observed. This pattern is consistent with the idea that countries perceived security pressures may experience more higher defence budget allocations. As in model 1, NATO membership, debt and energy intensity

variables are not statistically significant predictors. Lastly, the intercept if negative and statistically significant which indicates a baseline tendency towards over or under forecasting.

Taken together, these results suggest three main things. Firstly, forecasting and actual spending discrepancies in environmental and defence budgets are largely independent phenomena. Secondly, geographical proximity to Russia matters more for defence forecasting than for environmental forecasting. Thirdly, fiscal and alliance variables appear weakly related to forecasting accuracy.

4.2 Qualitative Results

Security proved to be a difficult-to-grasp, multifaceted concept. One distinction that profoundly shaped the understanding of security was the distinction between personal and collective security. As illustrated by the quote, *"I believe that security is actually often a collective matter, a societal question, and not really about how safe I feel personally"*, it was questioned whether personal security exists at all. Consequently, there is a collective entity to which security questions are assigned. Personal security was perceived as being constructed in a strong relation to external societal factors. This reveals tensions, given that feelings of insecurity and fears are rooted in individual lived experience: While reflecting on whether and how security can be constructed on a personal level, the experience of panic attacks at the onset of the Russian-Ukrainian war was also reported. Therefore, we can observe how individual experience is translated into larger-scale levels at which security issues emerge and through which security threats can be reasonably countered.

The political phenomena and developments at the larger-scale level were primarily driven by external factors, including the shift to the political right, migration, and, most prominently, the Russian-Ukrainian war. The shift to the political right was discussed mainly in relation to recent developments in Austria and Germany and their implications for social cohesion and attitudes toward climate protection. Migration was framed not as a threat to participants' own security, but as affecting climate and war refugees forced to leave their home countries.

Beyond the political issues raised regarding security, we examine the various actors and dimensions of security that emerge from the discussion and address them in the following sections. In general, when discussing security, we operate at the level of security perceptions: our research does not aim to define or analyse the current state of security and its threats per se, but to describe how these are perceived. We understand security perceptions as dynamic and shaped through mutual interaction between personal perceptions and collective meaning-making regarding complex and diverse security threats. Our focus group offers a glimpse into such a process: participants entered with a range of security perceptions, allowing for diverse perspectives on security issues, while also further negotiating these perceptions.

Actors

Several actors and their relationships proved crucial in addressing the security threats posed by climate change and war.

One relationship that is marked by significant tensions is that between the national government and society. The government was portrayed as an active actor, possessing the power to make budgetary decisions and the task to fulfil society's needs, while society was framed as rather passive and dependent on the government's budgetary and strategic decisions. This active-passive framing was reversed in the hypothetical scenario of war, with the society defending the country adopting a highly active role. We therefore observe an inconsistency in the active-passive relationship. Nevertheless, with respect to our central theme, the budget, the government is perceived as active rather than passive. In reaction, participants suggested more extensive forms of democratic involvement, such as grassroots democracy and citizens' assemblies.

Other concerns included the government's relative failure to fulfil a multitude of short- and long-term needs within a limited legislative period. *"And that, yes, that is not the question that the Federal Republic can ask itself at the moment, because at the moment it naturally has a society that it must satisfy at every possible level."* This led participants to question whether the government is well-equipped to address long-term challenges and resolve contradictions in addressing climate change and war. Hence, society is perceived as dependent on a government with limited capacity to effectively address and resolve the climate crisis and military conflicts.

Another interesting distinction between active and passive actors was drawn with respect to arms and the environment. The environment was portrayed as fragile and in need of protection from destructive forces, whereas the effect of arms on the environment was framed with a martial connotation: *"Arms are also something that attacks the environment"*. Furthermore, it was implied that acquiring arms would eventually result in their use. Consequently, the participants were highly critical of the government's intensified armament plans. During the discussion, however, the environment evolved from a rather passive to a more ambivalent actor, particularly in the context of climate change, with which it is deeply entangled. The environment was understood as a complex and fragile system that also entails important resources and serves as a means of subsistence for society. Climate change, by contrast, although there was a common consensus that the climate crisis is human-caused, was understood as an abstract, chronic, yet highly destructive force, particularly in the future. This resulted in a temporal distinction in which the environment is perceived as passive and in need of protection in the present, but active and destructive in the future.

The armaments industry emerged as another important actor, primarily associated with profit-driven interests and lobbying. Participants perceived the industry as exerting substantial influence on current politics, advocating levels of armament deemed excessive, or even to the extent that defence might shift into offence. Therefore, participants agreed that the armaments industry should not be further supported. Alongside distrust of the armaments industry, the media was also discussed as susceptible to lobbying and failing to

critically report on current armament developments. In the context of the Russian-Ukrainian war, it was emphasized that political elites of Russia were perceived as the enemy rather than the population itself. Society was conceived as being at the mercy of war decisions made by foreign political elites, accompanied by a strong sense of distance and inability to fully grasp them as a concrete threat. Other political actors discussed in this context were the institutions of the EU and NATO. Although participants critically questioned both, for instance, the EU's limited democracy and agency, the fact that Austria was embedded in a union of states and surrounded by NATO countries provided a certain sense of security.

Dimensions of Security

The dimensions of security are grounded in the conceptualization of personal and collective security and are closely intertwined with the actors described. Four dimensions of security were examined: security from war, climate security, supply security, and social security. Whereas security from war and climate security were induced by our questions, the participants added the dimensions of supply and social security.

Security from war was the dominant security dimension and reflected the participants' sense of security regarding current and potential future geopolitical conflicts. The spatial and temporal distance from the Russian-Ukrainian war, as well as the likelihood of war in Austria as a future scenario, strongly influenced this dimension. Other external factors to consider are the current shift to the right and participants' historical knowledge about fascist dictatorship, which fuelled insecurity. Security from war was rather understood as a collective sense of security. The actors arms, environment, and foreign governments played a key role in shaping perceptions of security in the context of war. Participants feared the potential use of arms, given the existing global armament. They established a direct link between the use of arms and resulting environmental destruction, a link they also fear. Another fear was the unpredictability of foreign governments' decisions, as illustrated in the following quotation: *"I would take the fighter jet, for a similar reason, because it's just one of those situations that can't be assessed personally. [...] When a fighter jet flies overhead and drops a bomb, that's someone else's decision"*. Similarly, during our photo activity, the picture of fighter jets was selected as the most threatening. Overall, differing assessments of current geopolitical conflicts led to divergent perceptions on the degree of security from war.

Climate security refers to participants' sense of security regarding current and future climate catastrophes. The ambivalent character of the environment as an actor is reflected in perceptions of threat from climate change. Consequently, participants judged the environment and its related climate security as a higher threat over time. Furthermore, differing views on the predictability of climate change and, hence, on climate security were examined. On the one hand, the complexity of the environment and potential tipping points render climate change unpredictable and threatening. On the other hand, participants *"know what can and will happen with climate change"*. Thus, climate change appears to be predictable and therefore less frightening. Although participants differed in their views on the predictability of climate change consequences, greater predictability was generally associated

with a greater sense of control and security. Climate security, or securing environmental sustainability, was understood as a means of achieving a socio-ecological sustainable way of living that would support security across the other dimensions of security.

Social security is characterized by the accessibility of basic services. This dimension is closely related to all other security dimensions. It was especially often linked to climate security, as environmental sustainability should be built on a functioning and guaranteed social provisioning system.

Supply security refers to a participant's sense of availability and affordability of essential resources. Perceptions of supply security were strongly shaped by the Russian–Ukrainian war, including self-experienced crises, such as the energy provisioning crisis. In addition, future prospects of an emerging food-provisioning crisis driven by climate change were feared. Severe resource shortages were considered to be exacerbated by military conflicts, climate change, and their interplay. In contrast to security from war, supply security was also framed as a personal security issue, as it is associated with households' financial circumstances. The participants wished to constrain global supply chains and develop standards for affordable and local food to mitigate supply insecurity.

Reflecting on all examined dimensions of security, a latent sense of losing control emerges. *"Well, I'd say we might as well start shooting at each other right now, then the issue would be settled."* This quote demonstrates the participants' feeling of being overwhelmed during the examination of security issues. Whether the distances from and probabilities of war, the complexity and (un)predictability of climate, or their possible negative effects on supply security, all underlie this sense of uncertainty about what to expect, which actors to trust, and what the future will bring. Alternatively, their sense of insecurity was alleviated by the prospect of local socio-ecological provisioning, which connects social, supply, and climate security as a foundation for secure living.

Although none of the security dimensions were directly linked to their corresponding budgets, reflections on these dimensions will yield deeper insights.

Budget

Budget is the central theme of our research, more specifically, the federal government's budget decisions. We introduced the defence and climate budgets to the focus group, and participants sought a social budget as a third main budget category.

In accordance with the sense of security from war, the defence budget decisions were perceived as closely connected to the Russian-Ukrainian war. Given the present government's political composition and geopolitical context, investments in the defence budget were perceived as prioritized. Participants criticized the quantity of the budget and judged it harmful to the economy and the environment: *"If I continue to ruin the economy and all the money is invested in machinery, and machinery is also something that attacks the environment"*. Based on the focus groups' reflections, defence expenditures should be allocated to increased diplomatic efforts, academic research, civil protection and weapons intended solely for the country's defence. The purpose of the defence budget should be to

prevent further escalation of current geopolitical tensions. In a utopian world, participants see no need for any defence budget. Yet, in an era of rising geopolitical tensions and distrust of foreign governments, there is a need for a relatively small and purposeful defence budget.

The climate budget refers to expenditures whose overall purpose is to mitigate and adapt to the climate crisis. Participants judged the government's climate budget decisions to be an inadequate response to the current degree of climate change and a distorted perception of its delayed, destructive effects. No further increase in the climate budget was expected, although a larger climate budget was judged particularly necessary for long-term security. The participants identified a lack of investment in renewable energy and public transport. Overall, participants wished to enable socially just climate protection through a purposefully used climate budget.

The social budget was described as a fund dedicated to addressing social inequality. It was introduced by our participants, because they were concerned that the increase in the defence budget is enabled by cutbacks in the social sectors. This is why they highlighted the need for social expenditures to invest in well-functioning, sustainable, and affordable education, as well as other provisioning systems in the country. Moreover, participants addressed the need to invest in fully transparent emergency plans to safeguard social and financial security in military or climate catastrophes.

The quantity of budgets is decisive, but the intended purpose of the expenditures was considered more fundamental. Therefore, security perceptions were primarily not linked to the amount of spending in each budget, but to the purposes for which the expenditures are used. A lack of confidence in the government's budgetary approach was evident. According to the participants, fundamental social provisions cannot be cut, and decisions should be guided by long-term perspectives.

Hierarchy of budget allocations in combination with the dimensions of security

"But of course the paradox is: which country am I supposed to defend if it no longer exists tomorrow?"

For the final task of our focus group, we asked the participants to collectively allocate the available expenditures among the defence, climate, and the added social budgets. The budget allocation was made considering perceived tight money and the resulting austerity measures.

Participants judged the climate and social budgets to be equally important and substantially higher than the defence budget (see Appendix). From a long-term perspective, participants emphasized the need for increased allocations to climate and social budgets. As described above, social, and climate security are regarded as foundational and were treated accordingly in participants' budget allocations. In contrast, the defence budget was kept within appropriate limits to enhance participants' sense of security from war. However, the participants clarified that if the defence budget is too high or inappropriately used, it could lead to unacceptable increases in geopolitical tensions and conflicts, hence increasing insecurities concerning war.

In line with the associated actors, the climate budget was considered fragile and in need of protection, whereas the defence budget was perceived as irreversible and likely to continue incurring high maintenance costs in the coming decades. Participants connected climate and military security as they perceived climate change and militarization as interconnected, self-reinforcing mechanisms. Therefore, participants assessed increase in the defence budget would entail the use of resources and the delayed use of arms, with consequential environmental destruction. The climate budget, in contrast, was assessed as a potential mitigating factor in future geopolitical conflicts over resources. Consequently, allocating a substantial share of the climate budget to environmental restoration could reduce projected defence budgets. Yet, inappropriate use of the climate budget, for example in resource-intensive technologies, was seen as a potential further exacerbation of resource scarcity and conflicts, which consequently increase the need for defence budgets. Cuts to climate and social budgets, alongside increases in defence spending, were understood as reinforcing global insecurity and heighten the potential for conflict.

To conclude, the government, as a budgetary decision-making actor, was perceived as capable of shaping both current and future circumstances, including geopolitical conflicts, the climate crisis, and social inequalities. Past and current climate, social, and defence budget decisions were criticized by the participants. Their preferred hierarchy of budget allocation, placing climate and social budgets as equally important and substantially higher than the defence budget, was not reflected in the government's budgetary approach. Furthermore, participants proposed alternative budgetary priorities, including preventing further escalation, investing in renewable energy, and expanding education. Importantly, security perceptions were not primarily linked to the absolute level of spending in each budget, but to the purposes for which the expenditures are used.

5 Discussion

To analyse the dynamics between defence and environmental budget spending as well as public perceptions of security, the quantitative and qualitative results are combined below to provide a more comprehensive picture. Overall, the quantitative results helped fill gaps regarding past budget developments at the EU level, while the qualitative findings offered a more in-depth perspective on current security perceptions in Austria.

A key difference, and an advantage of combining both approaches, lies in the different timeframes on which they are based: whereas the quantitative part allowed us to understand past budget allocations and potential trade-offs, the qualitative part adopted a contemporary perspective on security perceptions and discussed future developments and expectations of budget allocations. Hence, the focus group's future outlook is based on more recent budget developments than those considered in the quantitative analysis. Important in that context at the EU-level is the fact that the *Readiness 2030* facilitates and encourages growth in the defence sector in the EU (European Commission, 2025b), as well as the decision of NATO members to allocate 5% of GDP to defence by 2035 (NATO, 2025a). In Austria, the defence budget increased by more than 90%, from €2.7 billion in 2022 to about €5.2 billion in 2026 (Bundesheer, 2025; Bundesministerium für Landesverteidigung, 2022). There are no clear indications regarding the future development of the climate budget.

In the quantitative EU results, we see a slight indication of a declining climate budget and a rising defence budget, but both are within the 80% confidence interval and thus not significant. Although the effects of EU and NATO policies are not yet visible in our data, these developments are expected to materialize in the coming years, and increased defence expenditure is already evident in Austria's national data as well as in NATO's internal statistics: From 2023 to 2025, the number of NATO members spending 2% of their GDP on defence rose from ten to 31 (NATO, 2025b). Moreover, NATO's goal of spending 5% of GDP within 2035 is a clear indication that the rapid growth in defence expenditure will continue (NATO, 2025b). In this sense, our quantitative data analysis was conducted too early to capture the forthcoming increase in defence spending. Regarding the environmental budgets, we do not know how the environmental budget will develop or whether it will be influenced by increased military spending or not.

Our quantitative results suggest that defence spending in the Baltic countries exceeds the levels predicted by the ARIMA forecast. Their budgets started to increase already after the invasion of Crimea in 2014. A few other countries, namely Denmark, Hungary, Poland and Sweden, had statistically significant results in the ARIMA forecast, all countries in relative proximity to Russia. These results are supported by our findings from the second OLS model, which show that proximity to Russia results in an underestimation of defence budget development.

In our focus group, the geographical distance to the Russian-Ukrainian war also emerged as one of the main factors influencing the participants' sense of security from war. Even though there were divergent perceptions of the actual threat in Austria, participants largely agreed that the danger and the sense of insecurity would be considerably higher in a

neighbouring country. Nevertheless, the participants in the focus group felt that a certain amount of defence expenditure in Austria is necessary due to the current geopolitical tensions. Therefore, we can say that geographical distance to Russia influences budget allocation (quantitative) as well as security perceptions (qualitative).

The security perceptions evident in our qualitative findings can be linked to the securitization theory outlined in the literature review, which describes how securitization, as a combination of threat design and management, is dependent on the public perceiving a security issue as important and threatening enough (Balzacq, 2016). This understanding of security threats as socially constructed also emerged in certain parts of the focus group discussion, where participants explained and criticized current armament developments, attributing them to the influence of the armaments industry on politics and the media's failure to report critically on these developments and 'convince' the public that armament to this extent is necessary. The focus group therefore appeared to demonstrate an awareness that the current threat of war is subject to social negotiations and viewed recent defence budget developments, with many countries substantially increasing or even doubling defence spending, as excessive.

In the budget allocation exercise and beyond, expectations of our focus groups diverged from past and current budget developments. Aside from that, it is notable that an additional social budget was introduced by the participants. If we had more time to conduct our mixed methods project, it would have been interesting to include the social budget as well in the quantitative part of our analysis and analyse its relation to climate and defence budget developments. Furthermore, participants feared trade-off effects with increasing defence budgets leading to a decrease in the social and climate budget. However, the quantitative part of the project did not deliver any significant results showing that the past defence budget developments resulted in decreasing environmental budgets.

Nonetheless, a major finding was that the security perceptions were not primarily linked to the budgets in general and especially not to the amount of spending in each budget. Instead, the intended purposes for which the expenditures are used emerged as a bigger factor in influencing security perceptions. That this is a well-founded concern has already been mentioned in the literature review, with Barrell et al. (2021) outlining how the efficiency of environmental expenditures varies considerably among EU member states. Concerning the defence budget, participants suggested a modified use of the expenditures, which consisted of increased diplomatic efforts, academic research, civil protection and weapons solely for the country's defence. This diverges from the current configuration of Austria's defence budget, where the major part is used for the military and not for civil protection or research and development (Eurostat, 2025).

Overall, the government's budget allocation seems to be at odds with the focus group's assessment, which appears to mirror 'individuals with a high future orientation' who tend to incorporate temporally distant considerations more strongly in their assessment of security threats and to rank environmental threats higher than is typically the case (Carmi & Bartal, 2013). The government's perceived focus on rather immediate threats aligns well with

the ambivalent character of the environment over time: As the environment is perceived as passive and in need of protection in the present, it is not surprising that this passive framing combined with the short-term oriented outlook of the government results in a 'passive' climate budget, paying for catastrophes as soon as they arise but not mitigating them in advance. This is striking as participants outlined that increasing climate security and achieving a socio-ecological sustainable way of living would also enhance security across the other dimensions, alleviating future (defence) budgets. This can be understood against the background of the participants perceiving climate change and militarization as interconnected, self-reinforcing mechanisms. This is supported by findings indicating that higher military expenditure results in increasing carbon emissions and therefore jeopardizes reaching climate targets (Dong et al., 2025). Concerning the interrelation between climate change and militarisation, the "1% principle" (Scanlan, 2018) argues for comprehensive responses to both risks. But recent developments, combined with NATO's 5% of GDP on defence and the *Readiness 2030* preparations, show that there are massive and growing differences in prioritisation and responses to climate change versus a potential armed conflict.

As described in the literature review, the general process of budget allocation must be understood as an economic and political process. The focus group captured both aspects, the economic through their discussion of current austerity measures and budgets as potentially competing. However, their perception of the political nature of budget-making diverged from the framing in the literature, where a major role is attributed to public opinions, influencing the direction of budgetary changes. Conversely, the participants perceived themselves and society in general as mainly passive and dependent on the government's budgetary and strategic decisions to fulfil society's needs. Together with their proposals for more extensive forms of democratic involvement, such as citizens' assemblies, it would have been interesting to delve deeper into questions of power and how much influence they perceive themselves to have within the current system.

In a world of polycrisis, the growing complexity and interconnectedness between crises create competition not only between budgetary sectors but also competition in which crisis is deemed most urgent by the public (Rakowski et al., 2025). Despite the public's influence on budget allocation, the expectations of our focus groups diverged from past and current budget developments and intended uses. Combined with their passive perception of themselves as being exposed to the government's budgetary decisions and the complexity of and competition between security threats, this results in insecurity and an underlying feeling of losing control.

5.1 Limitations and Further Considerations

This study is subject to a few limitations. Regarding the qualitative component, the analysis is limited as only one relatively homogeneous focus group was conducted, which is in part a consequence of the snowball sampling strategy employed. While focus groups provide in-

depth insights into perceptions, interpretations and experiences, the homogeneous composition of the group limited the range of perspectives which biased the results towards shared political values.

In addition, the qualitative analysis involved the translation of key concepts and excerpts from German to English, which introduces the risk of interpretative bias or loss of nuance. Certain terms and expressions are embedded in specific linguistic and institutional contexts, and their meanings may not be fully captured or transferable across languages. Although two people worked on the analysis, and the translation was done at the final stage of the analysis, this process remains a potential source of analytical bias. One reflection on our recruitment process that may inform similar future research concerns the framing of the focus group: we sought to mobilize participants, among other strategies, through a flyer highlighting the issues of militarization and climate change. Although the flyer was intentionally designed to be accessible and non-intimidating, it appeared in our conversations with potential participants that it may have deterred individuals who had not previously engaged extensively with at least one of these topics. Instead, a framing of the discussion during the recruitment process that is less directly tied to our research question and that more broadly emphasizes personal security perceptions in everyday life could prove beneficial.

On the quantitative side, the analysis is highly constrained by data availability, as the most recent comparable data are from 2023, providing only one year of observations following Russia's invasion of Ukraine. Additionally, while increases in defence spending are already visible in absolute terms, particularly after directives of NATO in 2025 to substantially increase defence budgets, this change is not yet reflected in harmonised expenditure data. As a result, the analysis underestimates the magnitude and structural implications of current shifts in public spending. Another limitation pertains to the choice of environmental protection data – due to feasibility reasons the chosen dataset of the Eurostat database does not accurately reflect all types of spending that would categorise environmental protection. Furthermore, insights from the focus group suggest that social spending play a crucial role in shaping perceptions of security, yet these dimensions are absent from previous literature and our own. Furthermore, out of feasibility reasons, we had a common ARIMA (p, d, q) for all 30 countries for defence and environmental budgets, making the ARIMA forecast less fitting and accurate for each individual country.

Future research would benefit from updated and longer time series as new data becomes available, enabling the study of medium- and long-term budgetary reallocations between defence, environmental and social spending. Lastly, further research could explicitly incorporate spending on welfare, health, housing into the analysis of security, allowing for a broader and more grounded understanding of how security is constructed and budgetary trade-offs during perceived security threat times.

6 Conclusion

The EU is entering into a period of renewed defence mobilisation while simultaneously still facing the long-term security consequences of climate change. While political debates and policies increasingly prioritize “defence readiness”, climate change continues to pose systemic risk to European societies.

Against this background, the aim of our work was to explore the relationship between spending patterns and security perceptions, through connecting traditional defence to non-traditional environmental security domains. Previous research into macro-level expenditure and micro-level security perceptions has been conducted separately without acknowledging the interconnections between the two topics. We therefore combined a quantitative time-series analysis of EU-wide expenditure with a thematic and hermeneutical analysis of qualitative evidence from Austria to examine how defence and environmental spending trajectories in the EU relate to public perceptions of security.

Evaluating our quantitative and qualitative findings together reveals several key dynamics: First, defence spending increases are not even across Europe, but are more volatile and prevalent in countries geographically closer to Russia. This indicates that proximity to armed conflict impacts fiscal responses. Such a pattern is also reflected in the Austrian focus group discussion, in which perceived insecurity was strongly linked to geographical distance from the Russian-Ukrainian war. Second, regardless of concerns about fiscal competition, our quantitative analysis finds no evidence of a direct trade-off between defence and environmental budgetary spending. In contrast, focus group participants nevertheless feared that increased military budgets could crowd out climate and especially social investments. Third, our qualitative results demonstrate that security is perceived as multidimensional, encompassing climate, social and supply security alongside traditional state-centric security, suggesting a broader multifaceted understanding of security than is reflected in budget structures. Finally, participants stressed that the purpose of expenditure matters more than absolute budget size, suggesting that increases in defence budget do not automatically translate to higher perceived security.

Altogether, our findings provide an answer to how defence and environmental budget allocations interact with public perceptions of security in the EU and Austria. Our study shows that defence spending responds strongly to geopolitical shocks, while environmental spending follows more distinct national trajectories. Public perceptions of security extend beyond military threats to include climate and social dimensions. Overall, this suggests that although budget allocations and security perceptions influence each other, they do not evolve in perfect alignment.

Our findings have several implications for current policy debate. They suggest that European security policy could benefit from a broader understanding of security, going beyond military preparedness alone. While defence spending increases may be in response to immediate geopolitical threats, public perceptions indicate that that long-term climate and social threats are equally important to societal security. If governments intend to maintain

trust in the federal budgetary process, budgetary decisions need to balance immediate defence requirements with expenditures addressing long-term environmental and social security. Although defence budget increases do not necessarily crowd out climate expenditure, governments should beware of tensions between different dimensions of security that might emerge under fiscal pressure.

As we approached two research fields that have largely developed separately by applying a mixed methods approach, this analysis should be understood as an exploratory contribution. The project offers an initial framework on how fiscal priorities and social understandings of security interact and offers multiple aspects of these interactions that could be more closely examined in the future. As defence spending across Europe – and globally – is expected to increase further in the next years, future research will be able to reassess these dynamics with updated data, extend the analysis to other continents, and include more dimensions likely to play an important role in the perception of security policies such as trust in democratic institutions.

The EU currently focuses on defence readiness in response to growing geopolitical tensions while the security consequences of climate change continue to unfold more slowly but no less intensely. Our research shows that citizens do not see these threats separately but understand security as shaped by a multitude of dimensions, among which are military and environmental developments. As the EU prepares for potential future conflicts while confronting the “horizontal threat” of security consequences of climate change, balancing fiscal priorities with broader public understandings of security will remain a challenge for policymakers.

References

Arnall, A. (2023). Climate change and security research: Conflict, securitisation and human agency. *PLOS Climate*, *2*(3), 1-11. https://doi.org/10.1371/journal.pclm.0000072

Austrian Federal Ministry of Finance. (2025). *Report on Effective Action to Correct the Excessive Deficit*. https://www.bmf.gv.at/en.html#q=Report%20on%20Effective%20Action%20to%20Correct%20the%20Excessive%20Deficit&pg=1&t=simple&po=&mi=

Balzacq, T., Léonard, S., & Ruzicka, J. (2015). 'Securitization' revisited: theory and cases. *International Relations*, *30*(4), 494–531. https://doi.org/10.1177/0047117815596590

Barrell, A., Dobrzanski, P., Bobowski, S., Siuda, K., & Chmielowiec, S. (2021). Efficiency of environmental protection expenditures in EU countries. *Energies*, *14*(24), 8443. https://doi.org/10.3390/en14248443

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Bryman, A. (2012). *Social research methods* (4th ed.). Oxford University Press.

Brzoska, M. (2009). The securitization of climate change and the power of conceptions of security. *Sicherheit & Frieden*, *27*(3), 137–145. https://doi.org/10.5771/0175-274x-2009-3-137

Budge, I., & Farlie, D. J. G. (1983). Party competition: selective emphasis or direct confrontation? An alternative view with data. *Cadmus - EUI Research Repository (European University Institute)*. https://hdl.handle.net/1814/41171

Bundesheer. (2025). Verteidigungsbudget 2025/26: Mission Vorwärts wird zur Mission Aufwärts. https://www.bundesheer.at/budget-2025

Bundesministerium für Landesverteidigung. (2022). 2022 — Das Jahr des Umdenkens. bmlv.gv.at. http://www.bmlv.gv.at/cms/artikel.php?ID=11678

Campbell, K. M., Gulledge, J., McNeill, J. R., Podesta, J., Ogden, P., Fuerth, L., Woolsey, R. J., Lennon, A. T., Smith, J., & Weitz, R. (2007). *The Age of Consequences: The foreign policy and national security implications of global climate change*. https://doi.org/10.21236/ada473826

Carmi, N., & Bartal, E. (2013). Perception of environmental threat in the shadow of war: The effect of future orientation. *Human and Ecological Risk Assessment an International Journal*, *20*(3), 872–886. https://doi.org/10.1080/10807039.2013.798217

Carmona, P., Stef, N., Jabeur, S. B., & Zaied, Y. B. (2023). Climate change and government policy: Fresh insights from complexity theory. *Journal of Environmental Management*, *338*, 117831. https://doi.org/10.1016/j.jenvman.2023.117831

Carter, J., Ondercin, H.L., Palmer, G. (2020). Guns, Butter, and Growth: The Consequences of Military Spending Reconsidered. *Political Research Quartlerly*. https://doi.org/10.1177/1065912919890417

Christie, E. H. The Demand for Military Expenditure in Europe: The Role of Fiscal Space in the Context of a Resurgent Russia. *Defence and Peace Economics*, *30*(1), 72-84. https://doi.org/10.1080/10242694.2017.1373542

Climate Policy Initiative. (2025). *Global Landscape of Climate Finance*. https://www.climatepolicyinitiative.org/publication/global-landscape-of-climate-finance-2025/

Coghlan A., (2026) *A little book of r for time series* (last read 06.02.2026) https://a-little-book-of-r-for-time-series.readthedocs.io/en/latest/src/timeseries.html#arima-models

Copernicus Climate Change Service. (2024, November 7). The year 2024 set to end up as the warmest on record. https://climate.copernicus.eu/year-2024-set-end-warmest-record

Creswell, J. W. (2009). *Research design: Qualitative, quantitative, and mixed methods approaches* (3rd ed.). Sage.

Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). Sage.

Dahal, D. R. (2024). The Shifting Geopolitics. *Journal of Political Science*, 185–201. https://doi.org/10.3126/jps.v24i1.62863

Dong, W., Ran, Q., Liu, F., Deng, R., Yang, J., Wang, K., Wang, X., Zheng, D., Li, C., Liang, W., Chou, J., Yuan, W., & Chen, D. (2025). Rising military spending jeopardizes climate targets. *Nature Communications*, *16*(1), 4766. https://doi.org/10.1038/s41467-025-59877-x

European Commission. (2025a). *Preserving Peace — Defence Readiness Roadmap 2030*. https://doi.org/10.1093/law-oeeul/e66.013.66

European Commission. (2025, December 23b). *Introducing the White Paper for European Defence and the ReArm Europe Plan – Readiness 2030*. https://defence-industry-space.ec.europa.eu/eu-defence-industry/white-paper-european-defence-readiness-2030_en#as-surge-in-defence-spending

European Commission. (n.d.). *Acting on defence to protect Europeans*. Retrieved January 8, 2025, from https://commission.europa.eu/topics/defence/future-european-defence_en

European Environment Agency (2025, October 2). *Environmental protection expenditure*. https://www.eea.europa.eu/en/analysis/indicators/environmental-protection-expenditure-8th-eap

Eurostat. (2025). General government expenditure by function (COFOG) [Data set]. https://ec.europa.eu/eurostat/databrowser/view/gov_10a_exp__custom_10236830/default/table?lang=en

Florișteanu, E. (2023). Military and Defense Expenditures in the Current Security Context: Concepts and Influencing Factors. *Scientific Bulletin*, *28*(2), 194-204. https://doi.org/10.2478/bsaft-2023-0019

Froschauer, U., & Lueger, M. (2020). *Das qualitative Interview: Zur Praxis interpretativer Analyse sozialer Systeme* (2nd edn). utb GmbH. https://doi.org/10.36198/9783838552804

Green-Pedersen, C., & Mortensen, P. B. (2009). Who sets the agenda and who responds to it in the Danish parliament? A new model of issue competition and agenda-setting. *European*

Journal of Political Research, *49*(2), 257–281. https://doi.org/10.1111/j.1475-6765.2009.01897.x

Günlük-Senesen, G. (2002). Budgetary trade-offs of security expenditures in Turkey. *Defence and Peace Economics*, *13*(5), 385–403. https://doi.org/10.1080/10242690213509

Halkos, G. E., & Paizanos, E. A. (2013). *The impact of government expenditure on the environment: An empirical investigation*. Munich Personal RePEc Archive. https://mpra.ub.uni-muenchen.de/39957/1/MPRA_paper_39957.pdf

Liboreiro, J. (2025, March 21). Brussel rebrands "Rearm Europe" after backlash from leaders of Italy and Spain. *Euronews*. https://www.euronews.com/my-europe/2025/03/21/brussels-confirms-rearm-europe-rebrand-after-backlash-from-italy-and-spain

Looney, R.E. (1986). *Military Expenditures in Latin America: Patterns of Budgetary Tradeoffs*.

Maizels, A. & Nissanke, M. K. (1986). The Determinants of Military Expenditure in Developing Countries. *World Development*, *14*(9), 1125-1140. https://doi.org/10.1016/0305-750X(86)90115-4

Lueger, M., & Meyer, R. E. (2007). Objektive Hermeneutik. In R. Buber & H. H. Holzmüller (Eds), *Qualitative Marktforschung* (pp. 173–187). Gabler. https://doi.org/10.1007/978-3-8349-9258-1_11

Miró, J. (2023). Responding to the global disorder: the EU's quest for open strategic autonomy. *Global Society*, *37*(3), 315-335. https://doi.org/10.1080/13600826.2022.2110042

Mortensen, P. B. (2010). Political Attention and Public Policy: A Study of How Agenda Setting Matters. *Scandinavian Political Studies*, *33*(4), 356-380. https://doi.org/10.1111/j.1467-9477.2010.00254.x

NATO (2025a, 18. December) Defence expenditures and NATO's 5% commitment, https://www.nato.int/en/what-we-do/introduction-to-nato/defence-expenditures-and-natos-5-commitment?selectedLocale=

NATO (2025b) Press Release, Defence Expenditure of NATO Countries (2014-2025) https://www.nato.int/content/dam/nato/webready/documents/finance/def-exp-2025-en.pdf

NATO (2024, 11. March). NATO countries, https://www.nato.int/cps/en/natohq/nato_countries.htm

Neuman, W. L. (2014). Social research methods: Qualitative and quantitative approaches (7th ed.). Pearson.

Reiter, C., & Sardadvar, K. (2025). Hermeneutics as a method of interpretation: Line by line to the reconstruction of meaning. In M. Kusenbach & M. Pfadenhauer (Eds), Handbook of Interpretive Research Methods in the Social Sciences (pp. 242–258). Edward Elgar Publishing. https://doi.org/10.4337/9781803926391.00024

Rokicki, T., Ochnio, L., Koszela, G., Żak, A., Szczepaniuk, E. K., Szczepaniuk, H., Michalski, K., & Perkowska, A. (2019). Public expenditure on environmental protection in the European Union countries. *Rocznik Ochrona Środowiska*.

http://yadda.icm.edu.pl/yadda/element/bwmeta1.element.baztech-a0d949eb-d3f8-4cfe-9673-1d9d5fd0c171

Scanlan, D. O. (2018). *A Tale of Two Puzzles: Accounting for military and climate change expenditures*. Oxford Research Group.

Schattschneider, E. E. (1960). *The Semi-sovereign People*. New York: Holt, Rinehart & Winston.

Schuknecht, L., Moutot, P., Rother, P., & Stark, J. (2011). The Stability and Growth Pact: Crisis and Reform. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1791598

Shumway, R. H. & Stoffer, D. H. (2016) *Time Series Analysis and Its Applications - With R Examples* (4th ed), Springer Texts in Statistics

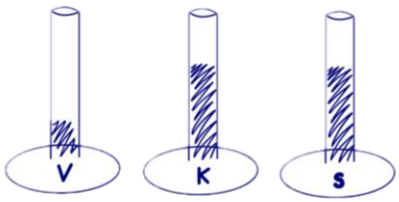
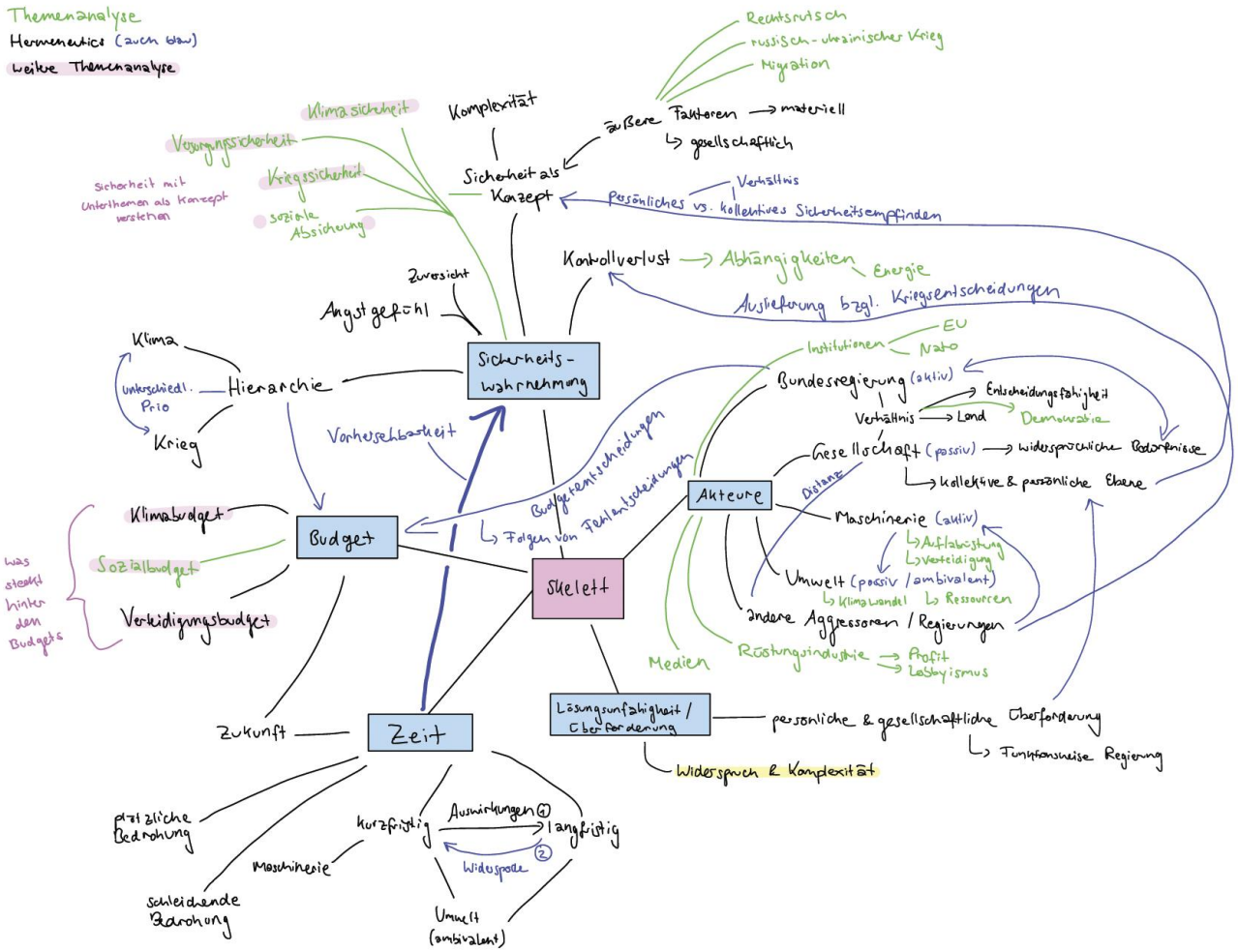
Stockholm International Peace Research Institute (SIPRI). (2025, April 28). *Unprecedented rise in global military expenditure as European and Middle East spending surges*. https://www.sipri.org/media/press-release/2025/unprecedented-rise-global-military-expenditure-european-and-middle-east-spending-surges

Tresch, R. W. (2023). *Public Finance: A Normative Theory* (4th ed.). London: Elsevier.

Wildavsky, A. (1964). *The Politics of the Budgetary Process*. Boston: Little, Brown and Company.

Appendix

Recreation of the collective budget allocation result in the focus group



Mind map combining hermeneutic and thematic analysis, main themes and connections